Roll No. ....

TCS–781

B. TECH. (IT)/B. TECH. (CSE-SE) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

SOFTWARE QUALITY ASSURANCE AND RELIABILITY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks

1. (a) Write a note on five views of software quality. (CO1)

OR

(b) Write a note on McCall's quality factors. (CO1)

2. (a) What are the six broad, independent categories of quality characteristics ? (CO2, CO3)

OR

(b) Application developer-A is at a company that is building their own book-lending website. Developer-A just picked up a story from the Kanban board to add a new feature that requires input from the user. Since Developer-A is a back-end developer, Developer-A must coordinate with the front-end team to add new data fields to the user interface. Developer-A opens an issue on the front-end team's GitHub to add the required fields. His feature requires a schema change (that is, a change to the table structure in the database) to hold the new data. So, Developer-A opens a ticket with the database team to make the change.

Developer-A works on the application logic for his feature and checks the code into the development branch in GitHub. Unfortunately, Developer-A can't test his code until the front-end team updates the user interface (UI) and the database team updates the schema. Because Developer-A is blocked Developer-A begins working on another feature story.

Two weeks later, the front-end team and the database team made their changes. Developer-A stops working on the feature Developer-A is currently working on to go back and merge those changes with his previous code. His attempt to merge encounters several merger conflicts. It takes several hours to resolve the merger conflicts because the code has drifted over time. Developer-A commits his updated code to the development branch. At the end of the month, the development team merges the development branch back into the master branch to prepare for release. Unfortunately, some quick bug fixes that were applied to the master branch now conflict with Developer-A's code in the development branch. Developer-A must spend an entire day remediating those conflicts.

The development team gets everything working and they are ready to deploy to production. They manually deploy the new release into production only to have it fail. The team had to roll back the release so they could determine the cause of the failure. They meticulously went through every feature change to find the cause of the failure. It took three days to test each new feature in isolation. They determined that the schema update for Developer-A's new feature was never applied. Once the schema was updated, everything started working. Answer the following questions with justification for your answer : (CO2, CO3)

(i) What could the company do differently to avoid having to spend time coordinating across teams ?

(ii) What could have been done to avoid the missing schema change ?

(iii) Should the development team have merged the development branch back into the master branch sooner ?

(iv) What could the development team have done differently to avoid the merger conflicts at the end of the month ?

3. (a) Assume a 1000 function point IT project such as an accounting package and 53,000 Java statements for alt cases. Assume 100 users of the software after deployment. Assume average complexity (problem, code, and data) in all four cases. The assumed test stages include (i) unit test, (ii) function test, (iii) regression test, (iv) component test, (v) usability test, (vi) performance test, (vii) system test, and (viii) acceptance test.

You are using Agile, Java, 10% reuse and static analysis but no inspect and no certified test personnel : (CO2, CO3)

Will the software delivered have high severity bugs ?

Will the software detivered have security flaws ?

Will the software delivered have error-prone modules ?

Is there a probability of "bad fixes" ?

What can be the average customer satisfaction ?

OR

(b) Associate the costs involved with software maintenance quality activities. (CO2, CO3)

4. (a) Explain the formal design review process with the help of process diagram. (CO4)

OR

(b) With a diagram, interpret SQA architecture. (CO4)

5. (a) Write a note on the classification of the causes of software errors. (CO5)

OR

(b) Point out the benefits and risks of involving external participants in the software maintenance process. (CO5)

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